

3.5 Classification of cross-sections

3.5.1 General

Cross-sections should be classified to determine whether local buckling influences their capacity, without calculating their local buckling resistance.

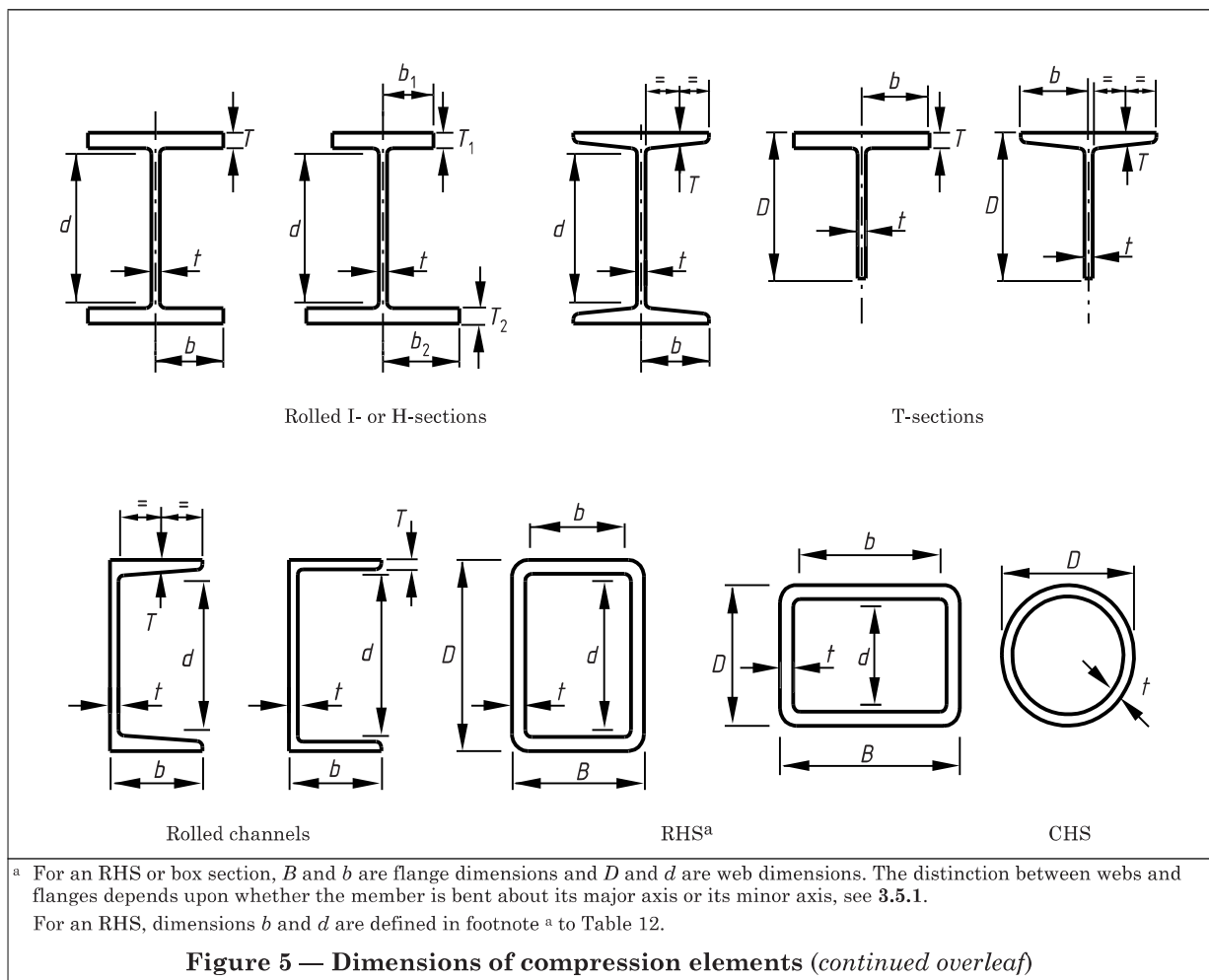
The classification of each element of a cross-section subject to compression (due to a bending moment or an axial force) should be based on its width-to-thickness ratio. The dimensions of these compression elements should be taken as shown in Figure 5. The elements of a cross-section are generally of constant thickness; for elements that taper in thickness the thickness specified in the relevant standard should be used.

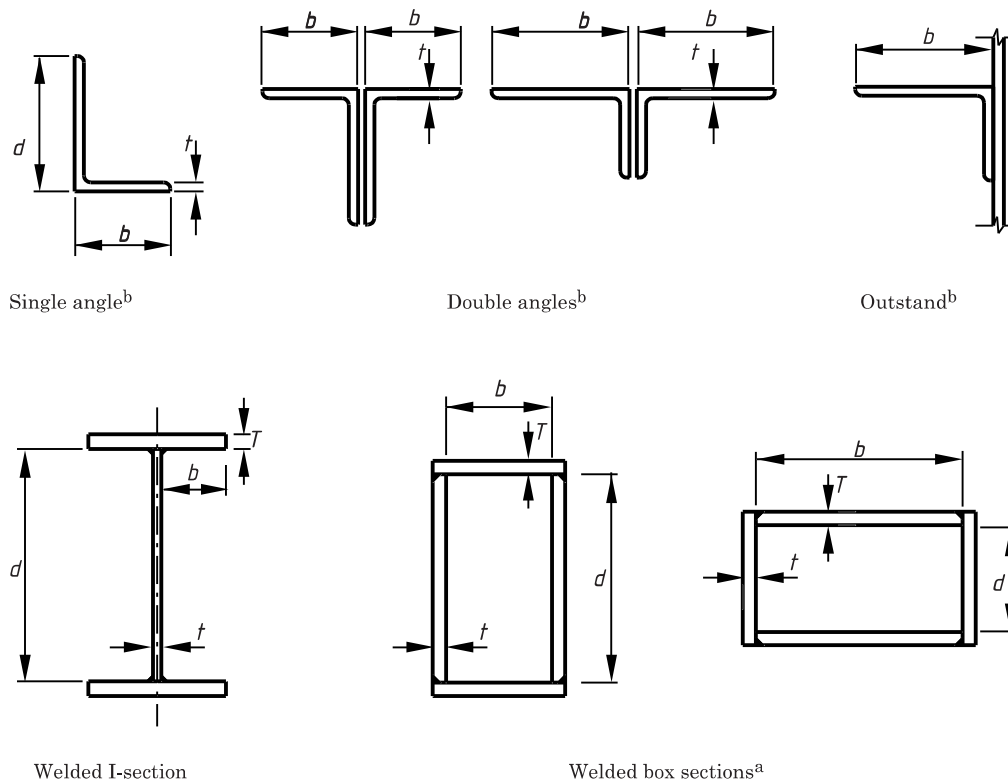
A distinction should be made between the following types of element:

- a) *outstand elements* attached to an adjacent element at one edge only, the other edge being free;
- b) *internal elements* attached to other elements on both longitudinal edges and including:
 - *webs* comprising internal elements perpendicular to the axis of bending;
 - *flanges* comprising internal elements parallel to the axis of bending.

All compression elements should be classified in accordance with 3.5.2. Generally, the complete cross-section should be classified according to the highest (least favourable) class of its compression elements. Alternatively, a cross-section may be classified with its compression flange and its web in different classes.

Circular hollow sections should be classified separately for axial compression and for bending.





^a For an RHS or box section, B and b are flange dimensions and D and d are web dimensions. The distinction between webs and flanges depends upon whether the member is bent about its major axis or its minor axis, see 3.5.1.

For an RHS, dimensions b and d are defined in footnote ^a to Table 12.

^b For an angle, b is the width of the outstand leg and d is the width of the connected leg.

Figure 5 — Dimensions of compression elements (continued)

3.5.2 Classification

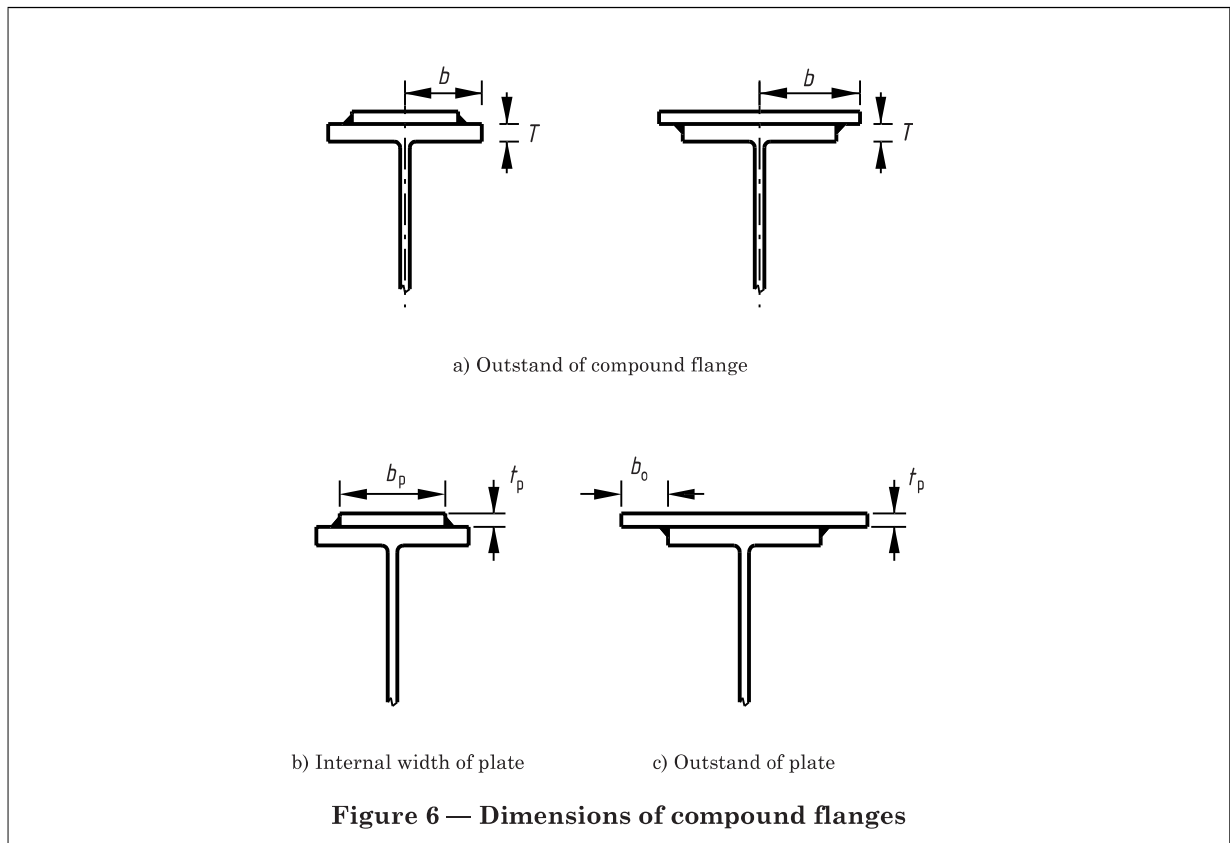
The following classification should be applied.

- Class 1 plastic: *Cross-sections with plastic hinge rotation capacity.* Elements subject to compression that meet the limits for class 1 given in Table 11 or Table 12 should be classified as class 1 plastic.
- Class 2 compact: *Cross-sections with plastic moment capacity.* Elements subject to compression that meet the limits for class 2 given in Table 11 or Table 12 should be classified as class 2 compact.
- Class 3 semi-compact: *Cross-sections in which the stress at the extreme compression fibre can reach the design strength, but the plastic moment capacity cannot be developed.* Elements subject to compression that meet the limits for class 3 given in Table 11 or Table 12 should be classified as class 3 semi-compact.
- Class 4 slender: *Cross-sections in which it is necessary to make explicit allowance for the effects of local buckling.* Elements subject to compression that do not meet the limits for class 3 semi-compact given in Table 11 or Table 12 should be classified as class 4 slender.

3.5.3 Flanges of compound I- or H-sections

The classification of the compression flange of a compound section, fabricated by welding a flange plate to a rolled $\overline{A_1}$ or welded $\overline{A_1}$ I- or H-section should take account of the width-to-thickness ratios shown in Figure 6 as follows:

- the ratio of the outstand b of the compound flange, see Figure 6a), to the thickness T of the original flange should be classified under “*outstand element of compression flange–rolled section*”, see Table 11;
- the ratio of the internal width b_p of the plate between the lines of welds or bolts connecting it to the original flange, see Figure 6b), to the thickness t_p of the plate should be classified under “*internal element of compression flange*”, see Table 11;
- the ratio of the outstand b_o of the plate beyond the lines of welds or bolts connecting it to the original flange, see Figure 6c), to the thickness t_p of the plate should be classified under “*outstand element of compression flange–welded section*”, see Table 11.



3.5.4 Longitudinally stiffened elements

For the design of compression elements with longitudinal stiffeners, reference should be made to BS 5400-3.

Table 11 — Limiting width-to-thickness ratios for sections other than CHS and RHS

Compression element			Ratio ^a	Limiting value ^b		
				Class 1 plastic	Class 2 compact	Class 3 semi-compact
Outstand element of compression flange		Rolled section	b/T	9ε	10ε	15ε
		Welded section	b/T	8ε	9ε	13ε
Internal element of compression flange		Compression due to bending	b/T	28ε	32ε	40ε
		Axial compression	b/T	Not applicable		
Web of an I-, H- or box section ^c	Neutral axis at mid-depth		d/t	80ε	100ε	120ε
	Generally ^d	If r_1 is negative:	d/t	$\frac{80\varepsilon}{1 + r_1}$ but $\geq 40\varepsilon$	$\frac{100\varepsilon}{1 + r_1}$	$\frac{120\varepsilon}{1 + 2r_2}$ but $\geq 40\varepsilon$
		If r_1 is positive:	d/t		$\frac{100\varepsilon}{1 + 1.5r_1}$ but $\geq 40\varepsilon$	
	Axial compression ^d		d/t	Not applicable		
	Web of a channel			d/t	40ε	40ε
Angle, compression due to bending (Both criteria should be satisfied)			b/t	9ε	10ε	15ε
			d/t	9ε	10ε	15ε
Single angle, or double angles with the components separated, axial compression (All three criteria should be satisfied)			b/t	Not applicable		15ε
			d/t			15ε
			$(b + d)/t$			24ε
Outstand leg of an angle in contact back-to-back in a double angle member			b/t	9ε	10ε	15ε
Outstand leg of an angle with its back in continuous contact with another component						
Stem of a T-section, rolled or cut from a rolled I- or H-section			D/t	8ε	9ε	18ε

^a Dimensions b , D , d , T and t are defined in Figure 5. For a box section b and T are flange dimensions and d and t are web dimensions, where the distinction between webs and flanges depends upon whether the box section is bent about its major axis or its minor axis, see 3.5.1.

^b The parameter $\varepsilon = (275/p_y)^{0.5}$.

^c For the web of a hybrid section ε should be based on the design strength p_{yf} of the flanges.

^d The stress ratios r_1 and r_2 are defined in 3.5.5.

Table 12 — Limiting width-to-thickness ratios for CHS and RHS

Compression element			Ratio ^a	Limiting value ^b		
				Class 1 plastic	Class 2 compact	Class 3 semi-compact
CHS	Compression due to bending		D/t	$40\varepsilon^2$	$50\varepsilon^2$	$140\varepsilon^2$
	Axial compression		D/t	Not applicable		$80\varepsilon^2$
HF RHS	Flange	Compression due to bending	b/t	28ε but $\leq 80\varepsilon - d/t$	32ε but $\leq 62\varepsilon - 0.5d/t$	40ε
		Axial compression	b/t	Not applicable		
	Web	Neutral axis at mid-depth	d/t	64ε	80ε	120ε
		Generally ^{cd}	d/t	$\frac{64\varepsilon}{1 + 0.6r_1}$ but $\geq 40\varepsilon$	$\frac{80\varepsilon}{1 + r_1}$ but $\geq 40\varepsilon$	$\frac{120\varepsilon}{1 + 2r_2}$ but $\geq 40\varepsilon$
		Axial compression ^d	d/t	Not applicable		
CF RHS	Flange	Compression due to bending	b/t	26ε but $\leq 72\varepsilon - d/t$	28ε but $\leq 54\varepsilon - 0.5d/t$	35ε
		Axial compression ^d	b/t	Not applicable		
	Web	Neutral axis at mid-depth	d/t	56ε	70ε	105ε
		Generally ^{cd}	d/t	$\frac{56\varepsilon}{1 + 0.6r_1}$ but $\geq 35\varepsilon$	$\frac{70\varepsilon}{1 + r_1}$ but $\geq 35\varepsilon$	$\frac{105\varepsilon}{1 + 2r_2}$ but $\geq 35\varepsilon$
		Axial compression ^d	d/t	Not applicable		

Abbreviations

CF Cold formed;

CHS Circular hollow section — including welded tube;

HF Hot finished;

RHS Rectangular hollow section — including square hollow section.

^a For an RHS, the dimensions b and d should be taken as follows:
 — for HF RHS to BS EN 10210: $b = B - 3t$; $d = D - 3t$
 — for CF RHS to BS EN 10219: $b = B - 5t$; $d = D - 5t$
 and B , D and t are defined in Figure 5. For an RHS subject to bending B and b are always flange dimensions and D and d are always web dimensions, but the definition of which sides of the RHS are webs and which are flanges changes according to the axis of bending, see 3.5.1.

^b The parameter $\varepsilon = (275/p_y)^{0.5}$.

^c For RHS subject to moments about both axes see H.3.

^d The stress ratios r_1 and r_2 are defined in 3.5.5.

3.5.5 Stress ratios for classification

The stress ratios r_1 and r_2 used in Table 11 and Table 12 should be determined from the following:

a) for I- or H-sections with equal flanges:

$$r_1 = \frac{F_c}{dtp_{yw}} \text{ but } -1 < r_1 \leq 1$$

$$r_2 = \frac{F_c}{A_g p_{yw}}$$

b) for I- or H-sections with unequal flanges:

$$r_1 = \frac{F_c}{dtp_{yw}} + \frac{(B_t T_t - B_c T_c) p_{yf}}{dtp_{yw}} \text{ but } -1 < r_1 \leq 1$$

$$r_2 = \frac{f_1 + f_2}{2p_{yw}}$$

c) for RHS or welded box sections with equal flanges:

$$r_1 = \frac{F_c}{2dtp_{yw}} \text{ but } -1 < r_1 \leq 1$$

$$r_2 = \frac{F_c}{A_g p_{yw}}$$

where

A_g is the gross cross-sectional area;

B_c is the width of the compression flange;

B_t is the width of the tension flange;

d is the web depth;

F_c is the axial compression (negative for tension);

f_1 is the maximum compressive stress in the web, see Figure 7;

f_2 is the minimum compressive stress in the web (negative for tension), see Figure 7;

p_{yf} is the design strength of the flanges;

p_{yw} is the design strength of the web (but $p_{yw} \leq p_{yf}$);

T_c is the thickness of the compression flange;

T_t is the thickness of the tension flange;

t is the web thickness.

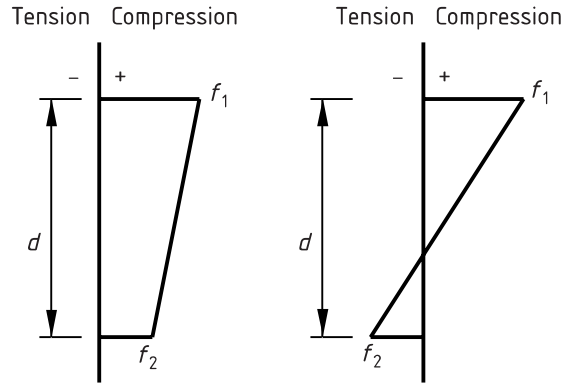


Figure 7 — Stress ratio for a semi-compact web

3.5.6 Effective plastic modulus

3.5.6.1 General

Class 3 semi-compact sections subject to bending should be designed using either the section modulus Z or the effective plastic modulus S_{eff} . For I- or H-sections with equal flanges, RHS and CHS, the effective plastic modulus should be determined from 3.5.6.2, 3.5.6.3 or 3.5.6.4 respectively. For I- or H-sections with unequal flanges subject to bending in the plane of the web, reference should be made to H.3. For other cross-sections S_{eff} should be taken as equal to the section modulus Z .

3.5.6.2 I- or H-sections with equal flanges

For class 3 semi-compact I- or H-sections with equal flanges, the effective plastic moduli $S_{x,\text{eff}}$ and $S_{y,\text{eff}}$ about the major and minor axes may be obtained from:

$$S_{x,\text{eff}} = Z_x + (S_x - Z_x) \left[\frac{\left(\frac{\beta_{3w}}{d/t} \right)^2 - 1}{\left(\frac{\beta_{3w}}{\beta_{2w}} \right)^2 - 1} \right] \quad \text{but} \quad S_{x,\text{eff}} \leq Z_x + (S_x - Z_x) \left[\frac{\frac{\beta_{3f}}{b/T} - 1}{\frac{\beta_{3f}}{\beta_{2f}} - 1} \right]$$

$$S_{y,\text{eff}} = Z_y + (S_y - Z_y) \left[\frac{\frac{\beta_{3f}}{b/T} - 1}{\frac{\beta_{3f}}{\beta_{2f}} - 1} \right] \quad \text{[A1]} \quad \text{but} \quad S_{y,\text{eff}} \leq S_y \quad \text{[A1]}$$

where

- b is the flange outstand, see Figure 5;
- d is the web depth;
- S_x is the plastic modulus about the major axis;
- S_y is the plastic modulus about the minor axis;
- T is the flange thickness;
- t is the web thickness;
- Z_x is the section modulus about the major axis;